

文献阅读笔记

安防机器人自动对接与充电

文献信息 Ren C. Luo, Chung T. Liao, Kuo L. Su, Kuei C. Lin. "Automatic Docking and Recharging System for Autonomous Security Robot." *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2005. DOI: 10.1109/IROS.2005.1545197。

一句话概括 论文通过“人工视觉地标 + 逆透视位姿估计 + 虚拟弹簧控制 + 两自由度被动柔顺接口”，构成从低电量判断到自动靠站、接触和安全充电的完整闭环，100 次试验机械对接成功率为 99%。

为什么这篇老论文仍值得读

它并不依赖复杂学习模型，却把感知、控制、机械容错、电气安全和电量预测串成一个可工作的系统。很多现代对接算法只讨论定位精度，而本文提醒我们：对接成功率往往来自“算法精度 × 机构容差 × 接触检测 × 异常处理”的乘积。

系统构成

- 安防机器人搭载 CCD、超声、红外、无线网络、电源传感器等；计算平台为 Pentium III 933 MHz/512 MB RAM。
- 充电站提供两组电源：12 V/最大 30 A 给计算机和传感器，36 V/最大 15 A 给电机及驱动。
- 机器人端采用锥形导入口，站端充电针由弹簧复位，形成 2 个被动自由度；允许最大约 5° 角度误差和 2 cm 水平偏差。
- 限位开关确认接触后等待 30 s 再上电；电流检测区分过流、无电流、正常充电和电池饱和，并执行报警/断电。

batteries for motors and their drivers. Under recharging, the computer is still alive in order to monitor the recharging schedule and depart from the docking station when recharging is completed.

The robot docking mechanism is shown in Fig. 3(a). They are mounted in front of the robot. The recharging adapters are at the inside of the holes. Each hole is shaped as a cone in order to help the docking smoothly. Since both the connection pins and adapters are rigid, the docking station is designed for providing compliance for reasonable robot docking angle and offset. When the robot is approaching to the docking station with offset and docking angle, the guiding stick mounted on the docking station functions to guide the recharging pins for inserting into the adapters. Fig. 4 shows the mechanism of docking station. Extension springs are used to bring the recharging pins back to the center after the robot disengages. A little entry offset and angle are allowed when robot is in the docking process. The maximum rotational angle and horizontal offset are 5 degree and 2 cm respectively.



Fig. 1 The security robot "Chung-Cheng Shin-Kong NO.1".

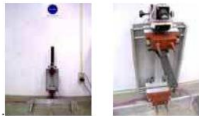


Fig. 2 The docking station

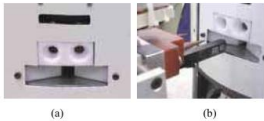


Fig. 3 The robot docking mechanism.

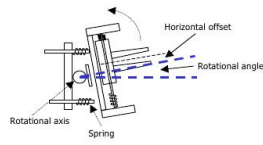
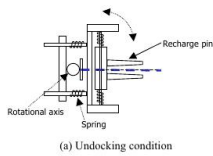


Fig. 4 Docking Device with 2 passive DOF.

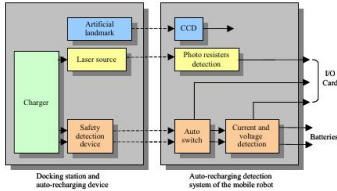


Fig. 5 The prototype of the auto-recharging device.

C. Auto-Recharging Device

The auto-recharging electrical arrangement schematic is shown in Fig. 6. The recharging flowchart in the docking station is shown in Fig. 7. The robot touches the limit switch that is mounted on docking station. The auto-recharging device can provide the power to the robot after 30 seconds. The current sensor can detect the recharging current value to prevent the faulty status. The current overload represents the condition of electrical system short. The battery saturation represents the batteries to be charged over. No current represents that the charger device has not attached to the security robot. Normal current represents that the security robot is charged with charger device. In the current overload and no current condition, the auto-recharging device must alarm and display faulty status. Then the recharging procedure must be stop. In the battery saturation condition, the recharging status delays 30 seconds. Then the auto-recharging device must cut off recharging current to the security robot.

The hardware block diagram of the auto-recharging device is shown in Fig. 8. The controller is microprocessor, and detects the recharging current using DC type current sensors. The input signals include limit switch and laser source. The output signals contain safety switch, laser light and display. The safety switch may be cut off or switched on the recharging current.



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图：原文第2页。算法之外的锥形导向、弹簧复位和双电压充电回路，是高功率的重要组成。

对接流程

1. 电量低于用户阈值后，利用已有导航系统返回充电站附近的初始点。
2. 机器人旋转搜索海军蓝色圆形人工地标，并通过相机变焦使其尽量占满视野。
3. 已知地标真实直径和安装高度，利用其在图像中的椭圆宽高做逆透视估计，得到距离与偏航角。

4. 先转向并平移到充电站正前方，再采用虚拟弹簧模型完成末端接近。
5. 接触后由限位开关与电流传感器接管充电安全逻辑。

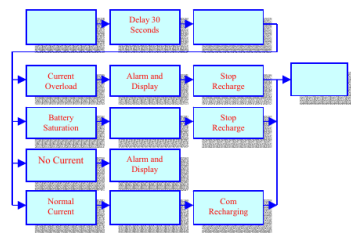


Fig. 7 The flowchart of the control system in auto-recharge process

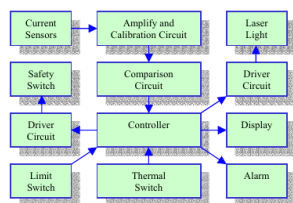


Fig. 8 The hardware block diagram of the auto-recharging device

D. Vision System Configuration

The configuration of vision system for docking is shown in Fig. 9. There is a camera mounted on top of the robot. The video signal is captured by the image frame grabber, which is installed inside the main controller. The acquired image is processed to detect the artificial landmark and locate the site of the docking station. Then the vision system guides the motion to achieve the docking. The motion commands are issued to the motion system through the inter-process communication (IPC). After the motion system executed the motion command, an acknowledge (ACK) command is sent back to the vision system in order to inform the further processing. The vision task is completed while the docking is successful.

In order to estimate the geometrical relationship between the robot and the docking station by inverse perspective projection, the height of the artificial landmark is setup as the same height as the camera, and right on top of the docking station. The height is known in advance. This landmark is specially designed for vision system. It is a navy blue circle. The diameter of this circle is also known.

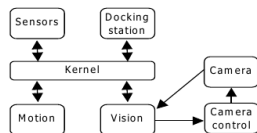


Fig. 9 Relationship between vision and robot system configuration.

III. DOCKING CONTROL

When the power is under the user-defined threshold, the robot should perform the following procedures:

- Step 1. Navigating the robot back to the position near the docking station.
- Step 2. Move the robot to the initial docking position.
- Step 3. Perform the docking by vision system.

The docking strategy is shown in Fig. 10. Robot is navigated to the initially docking position when the docking command is launched. The motion planning system navigates the robot back to the position A. There are three essential steps for vision in guiding the robot to achieve the docking.

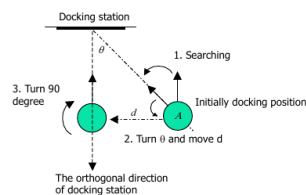


Fig. 10 Docking strategy.

- 1) Vision system searches and locks the artificial landmark.
- 2) Estimate the distance and angle in order to move robot to the front of the docking station.
- 3) Robot approaching control to dock into station by using virtual spring model.

The details are described as follows. First, vision system begins to search the artificial landmark by rotating the robot. The turning direction is predicted by the position estimation according to the internal dead reckoning. While the artificial landmark is detected, vision system issues a stop command to the motion system through IPC. The IPC is implemented by the method of sharing memory.

Second, camera zooming and robot turning are both employed to locate the landmark at the center of the image. The size of the landmark should be zooming as large as possible. It concerns the accuracy of the depth and direction angle from robot to the docking station.

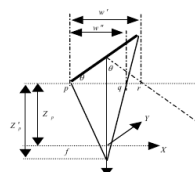


Fig. 11 Geometric relationship.

虚拟弹簧模型

作者把机器人与对接站想象成由一根可伸缩、可弯曲的弹簧连接。视觉系统给出距离 d 和角度 ϕ ：平移弹力与 d 成正比，弯曲恢复力与 ϕ 成正比；结合机器人质量、转动惯量和粘性阻尼，推导出线速度与角速度，再映射为左右轮速度。

这个模型的工程意义是把“位置误差”直接变成具有阻尼的连续控制量。与若干离散转向规则相比，它在末端接近时更平滑，也更容易与机械柔顺共同吸收小误差。

电量预测

论文采集电池在约 300 W 负载下的放电数据，以及最大 3.5 A 充电条件下的数据，采用最小二乘拟合电压—时间曲线。作者比较多项式阶次后认为三阶曲线对充放电过程较合适，并设置 11 V 阈值触发返航。

这里要注意：文中先完整推导了线性回归，再实际比较多阶多项式。真正可迁移的思想是“用在线采样估计剩余可用时间，并把返航能耗计入阈值”，而不是固定照搬 11 V。

主要结果

- 进行了 100 次循环试验：机器人离站、回到初始对接位置、视觉搜索、靠站、充电，再重复。
- 机械对接成功率 99%；作者把失败归因于地标区域特征量化误差。
- 图像序列显示机器人完成搜索、距离/方向估计、横向修正和末端靠站。
- 论文验证了充放电曲线拟合，但没有给出不同初始位姿下的时间、误差分布或置信区间。

$i=1,2,\dots,n$. ε_i and ε_j are not only uncorrelated but necessarily independent.

From the equation 14, we know that the response y_i in the i th trial is the sum of two components. One is the constant term $\beta_0 + \beta_1 x_i$ and the other is the random term ε_i . Therefore, y_i is a random variable. Considering simple linear regression model on y versus x is

$$y|x = E(y|x) + \varepsilon|x \quad (15)$$

We assume that

- 1) $E(y|x) = \beta_0 + \beta_1 x$
- 2) The error term $\varepsilon|x$ has mean $E(\varepsilon|x) = 0$, variance $Var(\varepsilon|x) = \sigma^2$ and independent of x .

Therefore, we can rewrite the linear regression model

$$y|x = \beta_0 + \beta_1 x_i + \varepsilon \quad (16)$$

In the paper, we use the least square estimation to estimate two parameters. First, we express the residual sum of square function to be

$$RSS(h_0, h_1) = \sum_{i=1}^n (y_i - h_0 - h_1 x_i)^2 \quad (17)$$

We want to find h_0 and h_1 to minimize $RSS(h_0, h_1)$.

Thus,

$$RSS(h_0, h_1) = \sum_{i=1}^n (y_i - \bar{y} - (h_0 - \bar{y} + h_1 \bar{x}) - h_1 (x_i - \bar{x}))^2$$

Let $a = h_0 + h_1 \bar{x} \Leftrightarrow h_0 = a - h_1 \bar{x}$

Then we rewrite $RSS(h_0, h_1)$ to $RSS(a, h_1)$

Therefore,

$$\begin{aligned} RSS(h_0, h_1) &= RSS(a, h_1) \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 + n(a - \bar{y})^2 + h_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 - 2h_1 \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) \end{aligned} \quad (18)$$

We define:

$$\begin{aligned} RXX &= \sum_{i=1}^n (x_i - \bar{x})^2 \\ RXY &= \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) \\ RYY &= \sum_{i=1}^n (y_i - \bar{y})^2 \end{aligned} \quad (19)$$

Thus, we rewrite (18)

$$RSS(a, h_1) = n(a - \bar{y})^2 + h_1^2 RXX - 2h_1 RXY + RYY \quad (20)$$

According to the definition of least square estimation, we have to minimize $RSS(h_0, h_1)$. Therefore,

$$\min_{a, h_1} RSS(a, h_1) = \min_{a, h_1} (h_1^2 RXX - 2h_1 RXY + RYY), a = \bar{y} \quad (21)$$

We want to find

$$h_1 = \frac{RXY}{RXX} \quad (22)$$

then the minimum value of $RSS(a, h_1)$ is

$$SYY(1 - \frac{RXY^2}{RXX * SYY}) \quad (23)$$

Therefore, we have a theorem here:

The ordinary least square estimate for simple linear regression is

$$\begin{aligned} \hat{\beta}_1 &= \frac{SXY}{RXX} \\ \hat{\beta}_2 &= \bar{y} - \hat{\beta}_1 \bar{x} \end{aligned} \quad (24)$$

The residual sum of square

$$\begin{aligned} RSS &= \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 * x_i))^2 \\ &= SYY(1 - \gamma^2(x, y)) \\ &= \sum_{i=1}^n \hat{\varepsilon}_i^2 \end{aligned} \quad (25)$$

V. EXPERIMENTAL RESULTS

The docking control algorithm and automatic recharging are implemented in our security robot. We employ a simple and recursive control scheme for recursively testing the docking algorithm and recharging device. The initial position of the robot is docked in the station. The control scheme controls the robot to depart from the station, and moves the robot to the initially docking position. Then the control scheme starts the vision algorithm for docking. After docking and recharging are finished, the control scheme performs the same work again. The successive frames in one trial are shown in Fig. 13. Only the key-point frames are captured from the video.

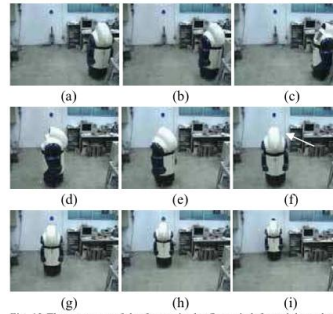


Fig. 13 The sequence of the frames in the figure is left-to-right and top-to-down. (a) The initial docking position. (b) Robot is in the searching state. (c) After the depth and direction of the docking station are estimated, the robot directly moves to the position by rotation and movement (frame c ~ f). Frame g to i exhibit the approaching to the docking station.

To determine the performance of the proposed method, 100 trials were performed. Experimental result shows a 99% success rate for mechanical docking. The successful rate of docking is high. We explain the reason of failure as the quantitation error for region features (a and b in formula 3).

We discharge the battery with a load about 300W (light bulb), and then acquire the related data with five-minute sampling time by the sensor data acquisition system. As the same way, we recharge the battery with no load by battery charger, which can provide 3.5 ampere-hour current (max). Then we can get n set observer data (x_i, y_i) for analysis. The diagram below shows the analysis result. In Case I, we will

图：原文第5页，关键帧从左到右、从上到下展示一次对接过程。

局限与批判性阅读

- 依赖尺寸、颜色和高度均已知的人工地标；照明变化、遮挡、地标污损与相机曝光会影响估计。
- “99%”是单一系统和固定实验环境的结果，论文未说明初始角度/距离的采样分布，难以判断工作域边界。

- 电池模型用电压—时间拟合，未显式纳入温度、老化、动态负载和回站路径能耗。
- 站端被动柔顺只有 5°和 2 cm 容差；它能兜底小误差，却不能替代全局定位与避障。

对今天系统设计的启发

现代实现可以把人工地标替换为 AprilTag、目标检测或点云几何，但应保留本文的三层容错思想：感知层把机器人送入捕获区，控制层平滑收敛，机械层吸收末端残差。充电端还应增加接触电阻/温升监测，并用 SOC/SOH 估计代替单一电压阈值。

来源与使用说明

本笔记依据本地保存的原论文整理；图片均为原文关键页面的本地渲染，仅用于团队内部学习、讨论和设计参考。笔记中的评价与启发属于阅读分析，不等同于作者原文结论。